

Student Takeaway for Sync Session

<< Week 8: Exploratory Data Analysis (EDA)>>

Designed to keep everything on track, it outlines key slide timings, essential talking points, and thought-provoking discussion questions. Stay focused on your objectives and ensure meaningful conversations, all within your available time.

Session Overview

Session title	Exploratory Data Analysis (EDA)								
Session duration	30 minutes								
Session type	Lectures: Concepts and techniques in EDA including handling missing values, outliers, encoding. Demonstrations: Hands-on EDA using Python and the Titanic dataset.								
Scope	This session introduces EDA and data preprocessing techniques, covering: - Understanding and treating missing values. - Identifying and removing outliers. - Encoding categorical features.								
Learning objectives	<table border="1"> <thead> <tr> <th>Objective</th><th>Core capability</th></tr> </thead> <tbody> <tr> <td>Understand the purpose of EDA and data cleaning</td><td>Analytical thinking in data preparation</td></tr> <tr> <td>Implement handling of missing values and outliers</td><td>Practical skills in Python</td></tr> <tr> <td>Apply encoding techniques</td><td>Understanding preprocessing needs</td></tr> </tbody> </table>	Objective	Core capability	Understand the purpose of EDA and data cleaning	Analytical thinking in data preparation	Implement handling of missing values and outliers	Practical skills in Python	Apply encoding techniques	Understanding preprocessing needs
Objective	Core capability								
Understand the purpose of EDA and data cleaning	Analytical thinking in data preparation								
Implement handling of missing values and outliers	Practical skills in Python								
Apply encoding techniques	Understanding preprocessing needs								
Software/Tools	- Python (Libraries: pandas, seaborn, matplotlib, sklearn) - IDE: Jupyter Notebook - Dataset: Titanic Dataset - Presentation Tool: PowerPoint								

Pro tips for success:

- Be Curious:** Ask why a step matters in preprocessing—it helps build intuition.
- Code Along:** Practice EDA in real-time as we demo.
- Engage:** Your observations during data exploration can unlock new insights!

Session Details

Slide(s)	Topic/Issue	What to expect	Notes
1	Introduction	Overview of EDA and session objectives	Why is EDA important in the ML pipeline?
2	Data Preprocessing	Steps like handling missing values, outliers, etc.	Set the foundation of clean, structured data

Slide(s)	Topic/Issue	What to expect	Notes
3	Handling Missing Data	Strategies using median/mode imputation	Practical example using Titanic dataset
4	Handling Outliers	Detect and treat using boxplots and IQR method	Identify "unusual" data entries
5	Feature Encoding	Convert text data to numeric using Label/One-Hot Encoding	Understand model requirements
6	Feature Scaling	Use StandardScaler and MinMaxScaler	Ensure numerical consistency
7	Feature Engineering	Create new features like family size and is_alone	Add meaning to data
8	Univariate Analysis	Plot distributions for continuous and categorical variables	Learn patterns from single-variable views
9	Bivariate Analysis	Explore relationships between features and target variable	Focus on survival prediction
10	Summary	Recap EDA steps and insights	Importance of EDA before model training
11,12	Q&A and Discussion	Open floor for clarifications and project planning	Get feedback and support

Post session

Reflection challenge	Which part of EDA surprised you the most—was it the impact of missing values, or how much feature engineering improved clarity?
Explore more	• Read: Kaggle kernels showing top EDA projects using Titanic data. • Watch: Videos demonstrating real-world EDA practices in ML competitions. • Play around: Apply the same EDA steps on new datasets like Iris, Wine, or Housing Price data.
Get inspired	Did you know that data preprocessing and EDA can improve model accuracy by over 30%? That's how powerful a good start is—mastering these skills will supercharge your ML journey.
The journey ahead	Next, explore Feature Selection and Dimensionality Reduction , where you'll learn how to remove redundant features, reduce noise, and enhance model efficiency using techniques like correlation analysis, PCA (Principal Component Analysis), and feature importance evaluation.